Linear Disentanglement in Representation Learning: Definition, Projection, and Knowledge Distillation

Technical Report

Abstract

Recent self-supervised vision models, notably DINO [Caron et al., 2021] and DINOv2 [Oquab et al., 2023], produce features with a remarkable property: they are *linearly separable* — a simple linear probe achieves near-supervised accuracy on ImageNet without any fine-tuning. Despite widespread empirical observation of this property, a precise mathematical characterisation and a principled method to *transfer* it to a student model via knowledge distillation remain underexplored. This report provides three contributions. First, we formalise *linear disentanglement* as a structural property of a feature distribution, decomposing it into two independently measurable axes: *isotropy* (eigenvalue uniformity of the covariance) and *decorrelation* (vanishing off-diagonal covariance). Second, we show that ZCA whitening is the canonical projection into the linearly disentangled space and characterise its optimality among orthogonal transformations. Third, we derive a three-term distillation objective — variance regularisation, covariance regularisation, and whitened-space alignment — that simultaneously *enforces* the structural property on the student and *transfers* semantic content from the teacher. We discuss connections to VICReg [Bardes et al., 2022], Barlow Twins [Zbontar et al., 2021], and Fisher Linear Discriminant Analysis.

Contents

1 Introduction **3**

2 Notation and Setup **3**

3 Defining Linear Disentanglement **3**
3.1 Classical Linear Separability vs. Representational Disentanglement 3
3.2 Isotropy: Eigenvalue Uniformity 4
3.3 Decorrelation: Off-Diagonal Covariance 4
3.4 Joint Characterisation 5

4 Transforming Features into the Linearly Disentangled Space **5**
4.1 ZCA Whitening 5
4.2 Optimality Among Whitening Transforms 6
4.3 Batch Approximation and Regularisation 6

5 Knowledge Distillation of Linear Disentanglement **7**
5.1 Problem Formulation 7
5.2 The Three-Term Objective 7
5.3 Gradient Analysis: Why Whitened Alignment Matters 8
5.4 Interaction Between the Three Terms 8
5.5 Integration with Existing Distillation Objectives 9

6 Multi-Teacher Linear Disentanglement Distillation **9**
6.1 Information Preservation under ZCA Whitening 9
6.2 Why Whitening is Especially Beneficial for MAE 10
6.3 Adaptive K Selection for Non-Isotropic Teachers 10
6.4 Teacher Suitability Analysis 12
6.5 Multi-Teacher LD Loss 12
6.6 Structural Effect on the Student Backbone 13
6.7 Practical Hyperparameter Guidance 13

7 Connection to Existing Methods **14**

8 Discussion **14**

9 Conclusion **15**

1 Introduction

Knowledge distillation typically trains a student network to mimic the *output* of a teacher network — matching class probabilities, intermediate activations, or pairwise relational structure. Such objectives transfer *what* the teacher knows but not necessarily *how* the teacher organises its internal representations.

DINO-style teachers are special: their patch tokens form rich spatial maps that can be directly clustered into semantically coherent segments without any supervision [Caron et al., 2021]. This is not merely a consequence of high accuracy; it follows from a structural property of the feature distribution. Concretely, DINO features are approximately *isotropic* and *decorrelated*, meaning the representation space is efficiently tiled by semantic directions with roughly equal energy. This property — which we formalise as *linear disentanglement* — is what makes linear probes succeed.

A student trained only to minimise cosine distance to teacher features will reproduce the directions of teacher features but may not reproduce their structural organisation. In particular, standard cosine or MSE losses are invariant to isotropic scaling and rotation: they do not penalise a student for concentrating all variance into a few directions (anisotropy) or for allowing correlated dimensions (entanglement).

This report addresses the question: *how can we distill not only the semantic content of a teacher but also its linear disentanglement property?*

2 Notation and Setup

Let $f_\theta : \mathcal{X} \to \mathbb{R}^D$ be a student vision encoder parameterised by θ. For a minibatch of N images $\{x_i\}_{i=1}^N$, define the feature matrix

$$Z \;=\; [f_\theta(x_1),\, \ldots,\, f_\theta(x_N)]^\top \;\in\; \mathbb{R}^{N \times D}.$$

For a ViT backbone, each image yields L patch tokens; we write $Z \in \mathbb{R}^{BL \times D}$ when the batch and spatial dimensions are flattened.

The centred feature matrix is $\tilde{Z} = Z - \mathbf{1}\bar{z}^\top$ where $\bar{z} = \frac{1}{N}\sum_i z_i$. The (unbiased) sample covariance is

$$C \;=\; \frac{1}{N-1}\,\tilde{Z}^\top \tilde{Z} \;\in\; \mathbb{R}^{D \times D}.$$

Let $C = U\Sigma U^\top$ be the eigendecomposition with $\Sigma = \mathrm{diag}(\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_D \geq 0)$.

A teacher encoder $g_\phi : \mathcal{X} \to \mathbb{R}^{D_t}$ is fixed and produces teacher features $Z^{(t)} \in \mathbb{R}^{N \times D_t}$. A projection head $h_\psi : \mathbb{R}^D \to \mathbb{R}^{D_t}$ maps student features into the teacher's dimension.

3 Defining Linear Disentanglement

3.1 Classical Linear Separability vs. Representational Disentanglement

The classical notion of linear separability states that a dataset $\{(z_i, y_i)\}$ is *linearly separable* if there exists a weight vector $w \in \mathbb{R}^D$ and bias b such that $w^\top z_i + b > 0$ whenever $y_i = +1$ and < 0 otherwise. This is a binary, label-dependent property.

For representation learning, we require a richer, label-free notion that characterises the *geometry* of the feature distribution. Intuitively, a representation is linearly disentangled if

1. it uses each dimension independently to encode distinct factors of variation (decorrelation), and

2. it allocates roughly equal energy to each dimension rather than collapsing into a low-dimensional subspace (isotropy).

These conditions do not by themselves guarantee that classes are separable — they constrain the unlabelled marginal distribution, not the class geometry (Section 3.4 makes this precise). Rather, they make a linear probe *efficient to fit*: (i) no dimension is redundant and (ii) no single direction dominates the gradient signal during probing, so the probe is well-conditioned and label-efficient whenever the classes are in fact separable.

3.2 Isotropy: Eigenvalue Uniformity

Definition 3.1 (Participation Ratio)**.** *The* participation ratio *(PR) of a feature matrix* $Z \in \mathbb{R}^{N \times D}$ *is*

$$\rho(Z) \;=\; \frac{\left(\sum_{i=1}^{r} \sigma_i\right)^2}{r \sum_{i=1}^{r} \sigma_i^2} \;\in\; \big(0,\, 1\big],$$

where the σ_i *are the eigenvalues of the sample covariance* C *(equivalently* $\sigma_i \propto s_i^2$ *for the singular values* s_i *of* $\tilde{Z}$*) and* $r = \operatorname{rank}(C)$*. Throughout, all spectral diagnostics (*ρ*, spectral entropy, and the effective rank of Section 6.3) are defined on this single eigenvalue (variance) spectrum, so they are mutually consistent.*

Proposition 3.2. $\rho(Z) = 1$ *if and only if* $\sigma_1 = \sigma_2 = \cdots = \sigma_r$ *(perfect isotropy).* $\rho(Z) = 1/r$ *if all variance is concentrated in a single direction.*

Proof. By the Cauchy-Schwarz inequality, $\left(\sum_i \sigma_i\right)^2 \le r \sum_i \sigma_i^2$ with equality iff all σ_i are equal. Dividing by $r \sum_i \sigma_i^2$ gives $\rho \le 1$ with $\rho = 1$ at equality. For the lower bound, when $\sigma_1 > 0$ and $\sigma_i = 0$ for $i > 1$, $\rho = \sigma_1^2/(r\sigma_1^2) = 1/r$. □

A related measure is the *spectral entropy*:

$$H(Z) \;=\; -\sum_{i=1}^{r} p_i \log p_i, \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}.$$

This is maximised at $H = \log r$ when all singular values are equal, recovering the uniform (isotropic) case. Both ρ and H are label-free diagnostics for collapse detection during training.

3.3 Decorrelation: Off-Diagonal Covariance

Definition 3.3 (Decorrelation Penalty)**.** *The* decorrelation penalty *of feature matrix* Z *is*

$$\mathcal{D}(Z) \;=\; \|C \odot (\boldsymbol{J} - \boldsymbol{I})\|_F^2 \;=\; \sum_{i \neq j} C_{ij}^2,$$

where $\boldsymbol{J}$ *is the all-ones matrix,* $\odot$ *is element-wise product, and* C *is the sample covariance.*

$\mathcal{D}(Z) = 0$ if and only if the feature dimensions are pairwise uncorrelated, i.e. C is diagonal. Combined with Definition 3.1, full linear disentanglement corresponds to $C = \sigma^2 I$ for some $\sigma > 0$.

Remark 3.4 (Information-Theoretic Interpretation)**.** *Minimising* $\mathcal{D}(Z)$ *is a second-order approximation to minimising the pairwise mutual information* $\sum_{i \neq j} I(Z_i; Z_j)$ *between feature dimensions. Under Gaussianity, the approximation is exact:* $I(Z_i; Z_j) = 0 \Leftrightarrow C_{ij} = 0$.

3.4 Joint Characterisation

Definition 3.5 (Linear Disentanglement)**.** *A feature distribution $p(z)$ with mean μ and covariance C is* (ρ_0, ε)-linearly disentangled *if*

$$\rho(Z) \;\geq\; \rho_0, \qquad \mathcal{D}(Z) \;\leq\; \varepsilon. \tag{1}$$

In the ideal limit $(\rho_0, \varepsilon) = (1, 0)$, the covariance reduces to a scaled identity: $C = \sigma^2 I$.

Connection to Linear Probe Performance. We stress that linear disentanglement is a property of the *marginal* (total) feature covariance C, which decomposes as $C = S_W + S_B$ into within- and between-class scatter S_W, S_B. Consequently $C = \sigma^2 I$ does *not* by itself imply class separability: it constrains only the sum $S_W + S_B$, not the class geometry, and one can have $C = \sigma^2 I$ with coincident class means (zero separability). Disentanglement should therefore be read as a *conditioning* property rather than a sufficient condition for a successful probe. Two precise statements make the connection rigorous.

(i) Whitening turns the optimal probe into nearest-class-mean. Under the homoscedastic-Gaussian model $z \mid y{=}c \sim \mathcal{N}(\mu_c, S_W)$ with shared within-class covariance S_W, the Bayes-optimal classifier is linear and ranks classes by the Mahalanobis distance $\|z - \mu_c\|_{S_W^{-1}}$. Applying the S_W-whitening $z \mapsto S_W^{-1/2} z$ makes this distance Euclidean, so the optimal probe reduces to nearest-class-mean and separability is governed by the *whitened* between-class separation $\left\| S_W^{-1/2}(\mu_a - \mu_b) \right\|_2$. This is the precise sense in which whitening is the best case for a linear probe — but it is whitening by S_W, not by the total covariance C, and it recovers exactly Fisher's criterion $J = \operatorname{tr}(S_W^{-1} S_B)$.

(ii) Isotropy improves probe conditioning. When the probe is fit by (regularised) least squares or gradient descent, its optimisation is governed by the condition number $\kappa(C) = \sigma_1/\sigma_D$. Isotropy ($\rho \to 1$, i.e. $\kappa \to 1$) minimises this condition number, giving faster, more label-efficient probe convergence and tighter margin-based generalisation bounds, *independently* of the class geometry.

The objective of Section 5 whitens by the total batch covariance — an unsupervised proxy that coincides with S_W only when $S_B \ll S_W$. It therefore delivers (ii) directly and approximates (i); a fully supervised S_W-whitening would be required to realise Fisher's discriminant exactly.

4 Transforming Features into the Linearly Disentangled Space

4.1 ZCA Whitening

Given features $Z \in \mathbb{R}^{N \times D}$, the goal is to find a linear transformation $W \in \mathbb{R}^{D \times D}$ such that the transformed features $Z_w = \tilde{Z} W$ satisfy $\operatorname{Cov}(Z_w) = I$.

Definition 4.1 (ZCA Whitening Transform)**.** *Let $C = U \Sigma U^\top$ be the eigendecomposition of the sample covariance. The* ZCA whitening matrix *is*

$$W_{\text{ZCA}} \;=\; U\, \Sigma^{-1/2} U^\top, \tag{2}$$

where $\Sigma^{-1/2} = \operatorname{diag}(\sigma_1^{-1/2}, \ldots, \sigma_D^{-1/2})$ (with regularisation $\sigma_i \leftarrow \sigma_i + \varepsilon$ in practice). The whitened features are $Z_w = \tilde{Z}\, W_{\text{ZCA}}$.

Proposition 4.2 (Covariance of Whitened Features)**.** $\operatorname{Cov}(Z_w) = I_D$.

Proof.

$$\mathrm{Cov}(Z_w) = W_{\mathrm{ZCA}}^\top C\, W_{\mathrm{ZCA}} = (U\Sigma^{-1/2}U^\top)^\top (U\Sigma U^\top)(U\Sigma^{-1/2}U^\top) = U\Sigma^{-1/2}\Sigma\Sigma^{-1/2}U^\top = I.$$

□

After ZCA whitening, the feature distribution achieves perfect linear disentanglement in the sense of Definition 3.5 with $(\rho_0, \varepsilon) = (1, 0)$. We caution that this is a statement about the marginal covariance only: when the spectrum is highly anisotropic, exact full-rank whitening amplifies near-zero directions and can *degrade* rather than improve probe accuracy. The regularised, rank-truncated variant of Section 6.3 is therefore what we use in practice; full-rank ZCA is the idealised limit, not the deployed transform.

4.2 Optimality Among Whitening Transforms

Many whitening transforms satisfy $\mathrm{Cov}(Z_w) = I$. The PCA whitening matrix $W_{\mathrm{PCA}} = U\Sigma^{-1/2}$ also achieves this but additionally rotates features into the eigenbasis of C. ZCA whitening has a unique optimality property:

Theorem 4.3 (ZCA Minimises Rotation)**.** *Among all whitening matrices W satisfying $W^\top C W = I$, the ZCA transform W_{ZCA} minimises the expected squared distance between original and whitened features:*

$$W_{\mathrm{ZCA}} = \arg\min_{W:\, W^\top C W = I} \mathbb{E}_z \left\| z - Wz \right\|_2^2 .$$

Proof. We write the per-sample whitening as $z \mapsto W^\top z$ (consistent with $Z_w = \tilde{Z}\, W$); since W_{ZCA} is symmetric this coincides with $z \mapsto Wz$ at the optimum. With z centred and $\mathbb{E}[zz^\top] = C$,

$$\mathbb{E}\left\| z - W^\top z \right\|^2 = \mathrm{tr}\big((I - W)^\top C\,(I - W)\big) = \mathrm{tr}(C) - 2\,\mathrm{tr}(CW) + \mathrm{tr}(W^\top C W).$$

Every whitening matrix satisfies $W^\top C W = I$, so $\mathrm{tr}(W^\top C W) = D$ is constant and minimising the distance is equivalent to *maximising* $\mathrm{tr}(CW)$. The constraint $W^\top C W = I$ is equivalent to $W = C^{-1/2}Q$ for some orthogonal Q, where $C^{-1/2} = U\Sigma^{-1/2}U^\top$ is the symmetric inverse square root (indeed $W^\top C W = Q^\top C^{-1/2} C\, C^{-1/2} Q = Q^\top Q = I$). Then

$$\mathrm{tr}(CW) = \mathrm{tr}\big(C\, C^{-1/2} Q\big) = \mathrm{tr}\big(C^{1/2} Q\big).$$

For the symmetric positive-definite matrix $C^{1/2}$, the orthogonal Procrustes inequality gives $\mathrm{tr}(C^{1/2}Q) \le \mathrm{tr}(C^{1/2})$ over all orthogonal Q, with equality iff $Q = I$. Hence the maximiser is $W = C^{-1/2} = U\Sigma^{-1/2}U^\top = W_{\mathrm{ZCA}}$. □

This optimality has a direct implication for distillation: applying ZCA whitening to teacher features produces a target that is as close as possible (in ℓ_2 sense) to the original teacher features while being perfectly linearly disentangled. The whitening does not arbitrarily rotate the semantic directions; it merely rescales them.

4.3 Batch Approximation and Regularisation

During training, the true covariance C is unknown and must be estimated from minibatches of size N. The estimator

$$\hat{C} = \frac{1}{N}\, \tilde{Z}^\top \tilde{Z}$$

is consistent but biased for small N. In practice two adjustments are needed.

Regularisation. Near-zero eigenvalues (dormant dimensions) cause numerical instability in $\Sigma^{-1/2}$. We replace σ_i by $\sigma_i + \varepsilon$ with $\varepsilon \in [10^{-4}, 10^{-2}]$. This also controls the *whitening radius*: strongly regularised whitening ($\varepsilon \gg \sigma_i$) approaches an isotropic identity transform, while $\varepsilon \to 0$ recovers exact ZCA.

Feature dimension. For high-dimensional features ($D \gg N$), the empirical covariance has rank at most N, and the pseudo-inverse should be used. In practice, for ViT patch features with $D = 768$ and $N = B \cdot L \geq 4096$, the batch is typically large enough for a well-conditioned estimate.

Computational cost. The dominant cost is the EVD (or SVD) of $\hat{C} \in \mathbb{R}^{D \times D}$, which is $\mathcal{O}(D^3)$ per step. For $D = 768$, this is approximately 4.5×10^8 FLOPs — comparable to one transformer attention layer. The whitening matrix W is applied as a single matmul $\tilde{Z}W$ at cost $\mathcal{O}(ND^2)$.

5 Knowledge Distillation of Linear Disentanglement

5.1 Problem Formulation

Standard feature-level distillation minimises

$$\mathcal{L}_{\text{distill}} = \mathbb{E}_x \Big[\ell\big(h_\psi(f_\theta(x)),\ g_\phi(x)\big) \Big],$$

where ℓ is typically cosine distance or MSE. This objective is invariant to arbitrary invertible linear transformations of the teacher's feature space, meaning the student may converge to a representation that has the same pairwise angles as the teacher but completely different covariance structure.

Our goal is to design $\mathcal{L}$ such that the student satisfies Definition 3.5 at convergence *without requiring class labels.*

5.2 The Three-Term Objective

Let $Z_s = h_\psi(f_\theta(\mathcal{B})) \in \mathbb{R}^{N \times D_t}$ be student features after the projection head on a minibatch $\mathcal{B}$, and $Z_t = g_\phi(\mathcal{B}) \in \mathbb{R}^{N \times D_t}$ be teacher features (stop-gradient). We propose:

$$\boxed{\mathcal{L}_{\text{LD}}(Z_s, Z_t) \ = \ \lambda_v \mathcal{L}_v(Z_s) \ + \ \lambda_c \mathcal{L}_c(Z_s) \ + \ \lambda_a \mathcal{L}_a(Z_s, Z_t)} \tag{3}$$

with the three terms defined as follows.

Variance term $\mathcal{L}_v$ — prevent dimensional collapse.

$$\mathcal{L}_v(Z_s) \ = \ \frac{1}{D_t} \sum_{j=1}^{D_t} \max\big(0,\ \gamma - \sqrt{\operatorname{Var}(Z_s^{(j)}) + \delta}\big), \tag{4}$$

where $Z_s^{(j)}$ denotes the j-th column of Z_s, $\gamma > 0$ is a target standard deviation (typically $\gamma = 1$), and $\delta > 0$ prevents gradient vanishing near zero. This hinge loss becomes active when any dimension collapses, pushing its variance back above γ.

Covariance term $\mathcal{L}_c$ — enforce decorrelation.

$$\mathcal{L}_c(Z_s) \ = \ \frac{1}{D_t} \mathcal{D}(Z_s) \ = \ \frac{1}{D_t} \sum_{i \neq j} \hat{C}_{ij}^2, \tag{5}$$

where $\hat{C}$ is the sample covariance of Z_s. This directly minimises the decorrelation penalty of Definition 3.3. Unlike the variance term, $\mathcal{L}_c$ is active at all times and enforces linear disentanglement as an ongoing structural constraint.

Whitened alignment term $\mathcal{L}_a$ — semantic transfer in the disentangled basis.

$$\mathcal{L}_a(Z_s, Z_t) \;=\; \frac{1}{N}\sum_{i=1}^{N}\big(2 - 2\,\langle \hat{z}^w_{s,i},\, \hat{z}^w_{t,i}\rangle\big), \tag{6}$$

where $\hat{z}^w = z^w / \|z^w\|$, and z^w denotes the ZCA-whitened, centred feature vector. The factor $2 - 2\cos(\cdot) \in [0, 4]$ equals the squared ℓ_2 distance on the unit sphere and provides stable gradients near convergence.

Remark 5.1. *The alignment term $\mathcal{L}_a$ differs from plain cosine distillation in two ways: (1) both student and teacher are whitened using their respective batch covariances before comparison, and (2) the whitening equalises the gradient magnitude across all feature dimensions (see Section 5.3).*

5.3 Gradient Analysis: Why Whitened Alignment Matters

Consider the gradient of the plain cosine alignment loss with respect to the student feature z_s (before any projection head):

$$\nabla_{z_s}\mathcal{L}_{\cos} = -\frac{1}{\|z_s\|}\Big(\hat{z}_t - \langle \hat{z}_s, \hat{z}_t\rangle \hat{z}_s\Big).$$

This gradient is proportional to the rejection of $\hat{z}_t$ from $\hat{z}_s$. In the original (non-whitened) space, dimensions with large variance in the teacher produce large gradient components, while low-variance dimensions are effectively ignored.

After ZCA whitening, the teacher features satisfy $\mathrm{Cov}(Z^w_t) = I$, so all dimensions have equal variance. The gradient of the whitened alignment loss

$$\nabla_{z_s}\mathcal{L}_a = W_s^\top\big(\hat{z}^w_t - \langle \hat{z}^w_s, \hat{z}^w_t\rangle \hat{z}^w_s\big),$$

where W_s is the student's batch whitening matrix, has magnitude spread uniformly across dimensions. This ensures the student must match *all* semantic directions in the teacher, not just the dominant principal components.

5.4 Interaction Between the Three Terms

The three terms are complementary and form a self-consistent system:

- $\mathcal{L}_c$ drives $C_s \to \mathrm{diag}(\sigma_1, \ldots, \sigma_D)$ (diagonal covariance).
- $\mathcal{L}_v$ drives $\sigma_j \geq \gamma$ for all j (no dimension is dead).
- $\mathcal{L}_a$ drives $Z^w_s \to Z^w_t$ (semantic alignment after whitening).

Together, $\mathcal{L}_v + \mathcal{L}_c$ enforce $C_s \approx \sigma^2 I$, i.e. the student features become linearly disentangled *as a structural property*, independently of any teacher. The alignment term $\mathcal{L}_a$ then provides a semantic *target* in the already-disentangled space.

Proposition 5.2 (Convergence Condition)**.** *Assume $\lambda_v, \lambda_c, \lambda_a > 0$ and that the student has sufficient capacity. At a stationary point of $\mathcal{L}_{\mathrm{LD}}$:*

1. $C_s \approx \sigma^2 I$ *(from* $\mathcal{L}_v + \mathcal{L}_c$*).*

2. *The student's whitened patch features are angularly aligned with the teacher's whitened patch features (from* $\mathcal{L}_a$*).*

3. *The student's isotropy score satisfies* $\rho(Z_s) \approx 1$ *(Definition 3.1).*

5.5 Integration with Existing Distillation Objectives

In a multi-teacher distillation framework, the total loss combines the LD objective with standard token-level and patch-level distillation:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{cos}}^{\text{cls}} + \mathcal{L}_{\text{rel}} + \mathcal{L}_{\text{cos}}^{\text{patch}}}_{\text{standard feature distillation}} + \underbrace{\mathcal{L}_{\text{LD}}}_{\text{linear disentanglement}} . \tag{7}$$

The LD term is applied selectively to teachers whose features exhibit linear disentanglement (DINO, DINOv2), while other teachers (CLIP, MAE) are handled by their standard distillation terms.

Crucially, the projection head h_ψ used in $\mathcal{L}_{\text{LD}}$ should be *separate* from the head used in $\mathcal{L}_{\text{cos}}^{\text{patch}}$. This is because: (i) cosine alignment is invariant to covariance structure and does not penalise entangled features; (ii) having a dedicated LD head allows $\mathcal{L}_{\text{LD}}$ to shape the representation without conflicting with the cosine supervision signal.

6 Multi-Teacher Linear Disentanglement Distillation

So far we have focused on distilling the linear disentanglement property from a single DINO teacher. A natural question arises: can we extend this framework to *all* teachers, including those (like MAE) whose features are not naturally linearly separable? And if so, what is gained by using multiple linearly-separable targets simultaneously?

6.1 Information Preservation under ZCA Whitening

The key observation is that ZCA whitening is an *invertible* linear map.

Proposition 6.1 (ZCA Inverse)**.** *The inverse of the ZCA whitening matrix* $W_{\text{ZCA}} = U\Sigma^{-1/2}U^\top$ *is*

$$W_{\text{ZCA}}^{-1} \;=\; U\,\Sigma^{1/2}U^\top.$$

Consequently, the whitened features $Z_w = \tilde{Z}\,W_{\text{ZCA}}$ *contain exactly the same information as the original centred features* $\tilde{Z}$*:*

$$\tilde{Z} = Z_w\,W_{\text{ZCA}}^{-1}.$$

This establishes that applying ZCA to *any* teacher — not only DINO — is a lossless operation. The whitened MAE features $Z_t^{(\text{mae}),w}$, for example, are in bijective correspondence with the original MAE encoder outputs. Distilling from $Z_t^{(\text{mae}),w}$ is therefore fully equivalent in information content to distilling from the raw features. This exact equivalence holds for *full-rank* ZCA; the rank-K truncation of Section 6.3 is a deliberate, lossy restriction to the teacher's signal subspace, trading the noise tail for numerical stability. Thus "lossless" refers to the full-rank map, whereas in practice we retain only the top-K informative directions.

6.2 Why Whitening is Especially Beneficial for MAE

Although ZCA preserves information for any teacher, the *practical benefit* of whitening is largest for teachers whose feature covariance is highly anisotropic — i.e., teachers for which $\sigma_1 \gg \sigma_D$. MAE is precisely such a teacher.

Spectral analysis of MAE features. MAE is trained to reconstruct masked image patches in pixel or frequency space. Its optimal encoder features tend to concentrate along directions that explain the most reconstruction variance — low-spatial-frequency components, colour statistics, and prominent textures. As a result, the empirical covariance of MAE patch tokens has a rapidly decaying eigenspectrum:

$$\sigma_1^{(\text{mae})} \gg \sigma_2^{(\text{mae})} \gg \cdots \gg \sigma_D^{(\text{mae})}.$$

By contrast, DINOv2's eigenspectrum is considerably flatter, consistent with its near-isotropic, linearly-separable representation [Oquab et al., 2023].

Gradient domination in raw-feature distillation. Consider an MSE loss between student projection Z_s and MAE teacher features Z_t in the eigenbasis U of $\text{Cov}(Z_t)$. The gradient of the loss with respect to the k-th principal component of the student is

$$\frac{\partial \mathcal{L}_{\text{MSE}}}{\partial [Z_s U]_k} \;=\; \frac{2}{N}\big([Z_s U]_k - [Z_t U]_k\big).$$

The mean squared magnitude of this gradient over the training distribution is $\mathcal{O}(\sigma_k^{(\text{mae})})$, since the teacher components have variance σ_k. Therefore, the update to the student is dominated by the $k = 1$ direction — the dominant reconstruction pattern — and the remaining $D - 1$ directions receive progressively weaker training signal.

Gradient equalisation after ZCA. After whitening, $\text{Cov}(Z_t^w) = I$, so every principal component of the whitened teacher has unit variance. The gradient magnitudes become uniform:

$$\mathbb{E}\left[\left(\frac{\partial \mathcal{L}_a}{\partial [Z_s^w]_k}\right)^2\right] \;=\; \mathcal{O}(1) \quad \forall\, k = 1, \ldots, D. \tag{8}$$

This forces the student to fit *all* of MAE's information — both the dominant reconstruction directions and the fine-grained, higher-frequency semantic details that raw MSE undersupervises.

6.3 Adaptive K Selection for Non-Isotropic Teachers

Full-rank ZCA whitening, as defined in Section 4.1, is well-motivated when the teacher's covariance matrix is well-conditioned (e.g. DINOv2). For teachers like MAE whose eigenspectrum decays rapidly, however, full-rank whitening causes two problems:

1. **Noise amplification.** Near-zero eigenvalues $\sigma_k \approx 0$ produce $\sigma_k^{-1/2} \gg 1$ in the whitening matrix, amplifying the directions that carry essentially no signal.
2. **Ill-conditioning.** Even with regularisation $\sigma_k \leftarrow \sigma_k + \varepsilon$, forcing the student to match noise directions wastes gradient steps and can destabilise training.

We therefore restrict both the whitening and the distillation objective to the *linearly separable subspace* of the teacher — the span of eigenvectors whose eigenvalues carry genuine signal. An adaptive criterion determines where signal ends and noise begins.

Effective rank as the primary criterion.

Definition 6.2 (Effective Rank)**.** *The* effective rank *of a feature matrix Z with centred covariance eigenvalues $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ is*

$$\operatorname{erank}(Z) \;=\; \frac{\left(\sum_{i=1}^{D} \sigma_i\right)^2}{\sum_{i=1}^{D} \sigma_i^2} \;\in\; [1,\, D]. \tag{9}$$

This is the participation-ratio effective rank; it satisfies $\operatorname{erank}(Z) = r \cdot \rho(Z)$ *with ρ and $r = \operatorname{rank}(C)$ as in Definition 3.1. We avoid the name "stable rank" for this quantity, as that term conventionally denotes the distinct operator-norm ratio $\|\cdot\|_F^2 / \|\cdot\|_2^2$.*

The effective rank equals D when all eigenvalues are identical (perfect isotropy) and equals r when only r eigenvalues are nonzero (rank-r matrix). It is continuous, differentiable, and bounded, making it a natural proxy for the effective dimensionality of the feature distribution.

Energy threshold as a secondary criterion. The energy-based rank is

$$K_{\text{energy}}(\tau) \;=\; \min\left\{ k : \frac{\sum_{i=1}^{k} \sigma_i}{\sum_{i=1}^{D} \sigma_i} \geq \tau \right\}, \tag{10}$$

for a threshold $\tau \in (0, 1)$ (typically $\tau = 0.90$). This guarantees that the selected subspace contains at least τ of the total feature variance, providing a signal-energy floor.

Combined adaptive criterion. We define the selected dimension as

$$\boxed{K \;=\; \operatorname{clip}\Big(\min\big(\lfloor \operatorname{erank}(Z_t) \rfloor,\; K_{\text{energy}}(\tau)\big),\; K_{\min},\; D\Big).} \tag{11}$$

Taking the *minimum* of the two criteria is conservative: a dimension is included only if *both* criteria consider it to carry signal. The floor of the effective rank provides one estimate of the active dimensionality; the energy threshold provides a complementary upper bound based on cumulative variance. Together they exclude the noise tail robustly, even when one criterion alone might be too permissive.

Table 1 reports typical K values for the two primary patch teachers.

Table 1: Typical adaptive K values for different teachers ($D = 768$, $\tau = 0.90$). The MAE values are approximate and depend on training dataset and model size.

Teacher	$\operatorname{erank}(Z_t)$	$K_{\text{energy}}(0.90)$	Selected K	K/D
DINOv2-B	≈ 700	≈ 680	≈ 680	88%
MAE-B	≈ 120	≈ 250	≈ 120	16%
MAE-L (D=1024)	≈ 180	≈ 350	≈ 180	18%

Top-K distillation in teacher's eigenbasis. Given the selected K and the corresponding top-K eigenvectors $U_k \in \mathbb{R}^{D \times K}$ of the teacher covariance, we modify the distillation procedure as follows.

1. **Project teacher:** $Z_t^{(k)} = \tilde{Z}_t\, U_k \in \mathbb{R}^{N \times K}$.
   Apply ZCA whitening to obtain $Z_t^{w(k)} = Z_t^{(k)}\, W_t$ where W_t is the $K \times K$ ZCA matrix of $Z_t^{(k)}$.

2. **Project student:** $Z_s^{(k)} = \tilde{Z}_s U_k \in \mathbb{R}^{N \times K}$ (gradients flow).
   Apply stop-gradient ZCA: $Z_s^{w(k)} = Z_s^{(k)} W_s^{\text{sg}}$ where W_s^{sg} is computed from $\text{sg}(Z_s^{(k)})$, so it is a constant w.r.t. gradient computation but the linear map $Z_s^{(k)} \mapsto Z_s^{w(k)}$ is differentiable.

3. **Losses (all in K-dim space):**

$$\mathcal{L}_v = \frac{1}{K} \sum_{j=1}^{K} \max(0,\, \gamma - \text{std}([Z_s^{(k)}]_j)), \tag{12}$$

$$\mathcal{L}_c = \frac{1}{K} \sum_{i \neq j} [\hat{C}_s^{(k)}]_{ij}^2, \tag{13}$$

$$\mathcal{L}_a = 2 - 2\, \frac{1}{N} \sum_i \langle \hat{z}_{s,i}^{w(k)},\, \hat{z}_{t,i}^{w(k)} \rangle. \tag{14}$$

Remark 6.3 (Stop-gradient whitening)**.** *The whitening matrix W_s^{sg} is computed from a stop-gradient copy of $Z_s^{(k)}$. This means the student's covariance is treated as a constant when computing gradients of $\mathcal{L}_a$, and the gradient of the alignment loss with respect to $Z_s^{(k)}$ is simply $W_s^{\text{sg}} \cdot \nabla_{\hat{z}} \mathcal{L}_a$. This avoids differentiating through the eigendecomposition (which is numerically fragile at repeated eigenvalues) while still allowing gradients to propagate to the student backbone through $Z_s^{(k)} = \tilde{Z}_s U_k$.*

Remark 6.4 (Why project both into teacher's basis, not student's?)**.** *One could instead project each of teacher and student into their* own *top-K eigenbases and align them separately. We prefer teacher's basis for two reasons. First, it ensures the student is trained to reproduce structure in the directions that are* meaningful for the teacher *— the informative signal space. Second, using separate eigenbases requires either a subspace alignment step (e.g. Procrustes rotation) or a CKA-style invariant loss, both of which reintroduce the covariance-invariance problem we are trying to solve.*

6.4 Teacher Suitability Analysis

Not all teachers in a multi-teacher framework benefit equally from LD distillation. Table 2 summarises the analysis.

Table 2: Suitability of each teacher for LD distillation.

Teacher	Features used	Token count N	Anisotropy	Apply LD?
DINOv2	Patch tokens	$B \times L$	Low (DINO naturally isotropic)	✓
MAE	Patch tokens	$B \times L$	High (reconstruction bias)	✓
CLIP	CLS token only	B	N/A (covariance ill-conditioned)	×
SAM	Patch tokens	$B \times L$	Medium (boundary-focused)	Optional

The critical factor is the token count N available per batch for estimating the covariance $\hat{C} \in \mathbb{R}^{D \times D}$. For patch-based teachers, $N = B \times L$ (e.g. $16 \times 196 = 3136 \gg D = 768$), giving a well-conditioned estimator. For CLIP, which provides only one global CLS token per image, $N = B \approx 32$–64, making the $D \times D$ covariance matrix severely rank-deficient and the ZCA transform numerically meaningless. LD distillation should therefore be applied to **patch-token teachers only**.

6.5 Multi-Teacher LD Loss

Let $\mathcal{T}_{\text{patch}} \subseteq \mathcal{T}$ denote the subset of teachers that provide patch tokens (e.g. $\mathcal{T}_{\text{patch}} = \{\text{dinov3}, \text{mae}\}$). For each teacher $k \in \mathcal{T}_{\text{patch}}$, we introduce a dedicated projection head

$h_k^{\mathrm{LD}} : \mathbb{R}^D \to \mathbb{R}^{D_k}$ that is *separate* from the standard patch-alignment head h_k. The multi-teacher LD loss is

$$\mathcal{L}_{\mathrm{LD}}^{\mathrm{total}} = \sum_{k \in \mathcal{T}_{\mathrm{patch}}} w_k \, \mathcal{L}_{\mathrm{LD}}\big(Z_s^{(k)},\, Z_t^{(k)}\big), \tag{15}$$

where $Z_s^{(k)} = h_k^{\mathrm{LD}}(f_\theta(\mathcal{B})) \in \mathbb{R}^{N \times D_k}$ and $Z_t^{(k)} = g_k(\mathcal{B}) \in \mathbb{R}^{N \times D_k}$ are the student and teacher patch features respectively, flattened over the spatial dimension. Each $\mathcal{L}_{\mathrm{LD}}$ applies the same three-term formula (Equation (3)) independently.

The total training objective becomes

$$\mathcal{L} = \underbrace{\sum_{k \in \mathcal{T}} \mathcal{L}_k^{\mathrm{std}}}_{\text{standard distillation}} + \underbrace{\mathcal{L}_{\mathrm{LD}}^{\mathrm{total}}}_{\text{multi-teacher LD}}, \tag{16}$$

where $\mathcal{L}_k^{\mathrm{std}}$ includes CLS cosine, relational, and patch alignment losses for teacher k.

Why separate LD heads? There are two reasons. First, the standard patch loss (cosine or MSE) is invariant to the covariance structure and can be satisfied by an anisotropic representation. Sharing weights between the standard and LD heads would create conflicting gradient signals. Second, having separate heads means the LD property is a *structural constraint on the backbone*: at inference time, the LD heads are discarded and the backbone itself produces linearly-separable features, directly usable by linear probes.

6.6 Structural Effect on the Student Backbone

We now describe informally what Equation (15) implies for the student backbone f_θ.

At a stationary point of $\mathcal{L}_{\mathrm{LD}}^{\mathrm{total}}$:

1. For each teacher k, the student projection $Z_s^{(k)}$ satisfies $C_s^{(k)} \approx \sigma^2 I$ (from the variance and covariance terms), meaning it is linearly disentangled with respect to teacher k's feature space.

2. $Z_s^{(k)}$ is angularly aligned with the ZCA-whitened teacher $Z_t^{(k),w}$ (from the alignment term), meaning the student captures all of teacher k's semantic content uniformly across eigendirections.

3. Since the above holds independently for *all* patch teachers, the backbone must produce representations that can simultaneously be projected into multiple linearly-disentangled spaces — each aligned with a different teacher's content.

This is a significantly stronger constraint than standard multi-teacher distillation: the student cannot satisfy it by collapsing to a low-dimensional subspace or by concentrating features along a single teacher's dominant directions. The multi-teacher LD objective creates an implicit pressure for the backbone to learn a *universally isotropic* representation that contains structured, linearly-separable information across multiple visual modalities (semantics, reconstruction, boundary detection).

6.7 Practical Hyperparameter Guidance

The loss weights $\{\lambda_v, \lambda_c, \lambda_a\}$ in Equation (3) and per-teacher weights w_k in Equation (15) require tuning. We offer the following guidance.

Relative scale of $\lambda_v : \lambda_c : \lambda_a$. Following VICReg [Bardes et al., 2022], we recommend $\lambda_v = \lambda_a = 25$ and $\lambda_c = 1$. The covariance term involves D^2 off-diagonal entries and thus

has a much larger raw magnitude; the factor of $1/D$ normalisation in $\mathcal{L}_c$ (Equation (5)) compensates for this.

Per-teacher weight w_k. We recommend setting $w_k = 1$ for all teachers initially. If the MAE LD loss converges faster than the DINO LD loss (because MAE's anisotropy is easily corrected by whitening), one may reduce w_{mae} after warm-up.

Whitening ε. Use $\varepsilon = 10^{-3}$ for both DINO and MAE. MAE's near-zero eigenvalues (dormant dimensions) are more common, so a slightly larger $\varepsilon = 5 \times 10^{-3}$ may improve stability for MAE.

7 Connection to Existing Methods

VICReg [Bardes et al., 2022]. VICReg applies variance, invariance, and covariance regularisation directly to *two augmented views* of the same image in a self-supervised setting. Our $\mathcal{L}_v$ and $\mathcal{L}_c$ are identical to VICReg's variance and covariance terms applied to the student. The key difference is that we replace the invariance term (which requires two views) with a whitened alignment term that leverages a pre-trained teacher — enabling the structural property to be acquired *via distillation* rather than self-supervised contrastive learning.

Barlow Twins [Zbontar et al., 2021]. Barlow Twins minimises $\|C_{YY} - I\|_F^2$ where C_{YY} is the normalised cross-correlation matrix between two views. This simultaneously drives diagonal entries to 1 (variance, our $\mathcal{L}_v$) and off-diagonal entries to 0 (decorrelation, our $\mathcal{L}_c$). We apply the same spirit but decompose the two terms, allowing independent weighting.

W-MSE [Ermolov et al., 2021]. Whitening Mean Squared Error (W-MSE) applies online whitening to feature patches before computing MSE across augmentations. Our $\mathcal{L}_a$ is conceptually similar but differs in that we whiten the *teacher's* representation — this transfers semantic structure from a frozen expert rather than enforcing invariance across augmentations.

CKA-based distillation. Linear CKA, defined as

$$\text{CKA}(X, Y) = \frac{\left\|X^\top Y\right\|_F^2}{\|X^\top X\|_F \, \|Y^\top Y\|_F},$$

is invariant to orthogonal transformations and to global isotropic scaling of either argument. It is, however, *not* invariant to arbitrary invertible linear maps — that property belongs to CCA, not CKA — so, contrary to a common shorthand, CKA is *not* whitening-invariant. Nevertheless CKA is unsuitable as an objective for our purpose. As a rotation- and scale-invariant *similarity index* it references no canonical basis and no absolute scale, so maximising $\text{CKA}(Z_s, Z_t)$ does not pin the student covariance to any prescribed structure such as $C_s = \sigma^2 I$, and it cannot distinguish a decorrelated student from an orthogonal rotation of it that scrambles the basis. CKA can therefore transfer relational structure but cannot *enforce* the disentanglement target, which is intrinsically basis- and scale-dependent; our LD objective explicitly addresses this gap.

8 Discussion

Why does DINO naturally produce linearly disentangled features? DINO's self-distillation objective, combined with the centering and sharpening operations in its student/teacher update, implicitly prevents collapse (addressing $\mathcal{L}_v$) and encourages diverse,

distributed representations across the full feature space. The centering operation in particular acts as a running estimate of feature mean, preventing any single direction from consistently dominating — a mechanism analogous to our $\mathcal{L}_c$.

Universality: applying LD loss to other teachers. The framework naturally extends to any teacher. For MAE features, which are good at dense prediction but not as isotropic as DINO, one could apply $\mathcal{L}_{\text{LD}}$ to distill a more isotropic version of MAE features. This would transfer MAE's spatial predictive power while adding the linear separability structure of DINO.

The role of the separate LD head. Using a dedicated projection head for the LD objective has an additional benefit: the head itself can be discarded at inference time. The student backbone f_θ is trained so that its intermediate features are linearly disentangled — the LD property is baked into the backbone, not the head. This is analogous to how DINO uses a separate projection head for its self-distillation but evaluates the backbone directly with linear probes.

Batch size sensitivity. The batch ZCA estimator (Section 4.3) requires $N \gg D$ for a well-conditioned covariance matrix. In practice, for $D = 768$ with $B \times L = 16 \times 196 = 3136$ tokens per batch, the covariance is well-estimated. However, for very small batches or large feature dimensions, a momentum-based running covariance estimate may be more stable.

9 Conclusion

We have formalised the linear disentanglement property through two complementary axes — isotropy (participation ratio) and decorrelation (off-diagonal covariance penalty) — and shown that $C = \sigma^2 I$ is the ideal covariance structure for linear probe performance.

ZCA whitening provides the canonical projection into the linearly disentangled space: it is the unique orthogonalisation that minimises the ℓ_2 distance from the original features while achieving $\text{Cov}(Z_w) = I$ exactly. Its optimality (Theorem 4.3) means that teacher semantics are preserved as closely as possible under the whitening constraint.

The three-term distillation objective $\mathcal{L}_{\text{LD}}$ uses the variance and covariance terms to impose the structural property on the student, and the whitened alignment term to transfer teacher semantics in the disentangled basis. This design bridges self-supervised structural regularisation (VICReg-style) and feature-level knowledge distillation (DINO-style) in a single, principled objective.

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